Software Requirements Specification

for

Food Saver System

Version 1.0

Prepared by CS Year 5, SE Old Section, G1

Imam Abdulrahman Bin Faisal University

College of Science and Humanities – Jubail

Department of Computer Science

31 October 2026

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Team Members

Table 1: Team members

Name	ID
Aljazi Alsaiari	2230004865
Roweena Belal	2230005940
Noor Alawami	2230000703
Forat Alabbas	2230001270

Revision History

Name	Date	Reason For Changes	Version
Food Saver Team (G1)	31 Oct 2026	First version of the SRS	1.0

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Food Saver application, version 1.0. It describes what the application shall do, who will use it, and the limits it has to work within. The SRS covers the whole application: account creation and log in, the food inventory, expiry alerts, recipe suggestions, food donation and requests, search and filter, donation matching, the recipe community and the rewards feature. It also lists the external interfaces, the nonfunctional requirements and the analysis models (use case and class diagrams) of the system. This SRS follows the Software Project Management Plan (SPMP) that was prepared earlier for the same project.

1.2 Document Conventions

This SRS follows the IEEE recommended practice for software requirements specifications (IEEE Std 830-1998) and the SRS template by Karl Wiegers. The following conventions are used in this document:

- Each functional requirement has a unique ID in the form REQ-n, and the numbering continues across all features in Section 4.
- The word “shall” is used to state a requirement that the system must meet.
- Every feature in Section 4 has its own priority (High, Medium or Low) and ratings for benefit, penalty, cost and risk on a scale from 1 (lowest) to 9 (highest). The detailed requirements of a feature take the same priority as the feature.
- Use case diagrams are drawn using UML notation, where «include» shows a step that always happens as part of a use case and «extend» shows an optional step.
- Tables and figures are numbered in order, and they are listed at the start of the document.
- Terms written as TBD are still to be decided, and they are collected in Appendix C.

1.3 Intended Audience and Reading Suggestions

This document is written for different readers, and each reader can focus on the parts that matter most to them:

- Developers and designers: to understand the features and requirements they need to build. They should read Sections 2, 3 and 4 in detail.
- Testers: to check that the application does what is written. Sections 4 and 5 are the most important for them, because each requirement can be tested.
- Project manager and team members: to follow the scope and track the progress. Sections 1, 2 and Appendix C are the most useful for them.
- Course instructor and reviewers: to evaluate the requirements and models. We suggest reading the whole document in order.

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- End users: to get an idea of what the application offers. Sections 1.4 and 2.2 give a simple overview.

The document is organized as follows: Section 1 introduces the product, Section 2 gives an overall description, Section 3 describes the external interfaces, Section 4 lists the system features with their use cases and functional requirements, Section 5 covers the nonfunctional requirements, and Section 6 covers other requirements. The appendices contain a glossary, the analysis models and the TBD list.

1.4 Product Scope

A large amount of food is thrown away every day at home because people forget what they have, or notice too late that it has expired. The UNEP Food Waste Index Report 2024 estimates that households produce most of the food waste in the world [4]. At the same time, many families in the same city need food. Food Saver is a mobile application that helps people reduce this waste in a simple way. Users record the food they have at home, and the application reminds them before items expire, suggests recipes that use these items, and lets them donate extra food to people nearby who need it.

The main goals of Food Saver are:

- Reduce household food waste by tracking food items and their expiry dates.
- Help users cook with what they already have through smart recipe suggestions.
- Connect people who have extra food with people who need it in the same area.
- Encourage users to keep donating by giving them points that can be turned into volunteer hours.

The version described in this SRS is a prototype built in Figma. It shows the full flow of the application using sample data only, so no real personal data or real locations are used. The requirements are still written as “the system shall …” because they describe how the final application is expected to behave.

The application has two roles. Table 2 shows the main functions of each role.

Table 2: Roles of Food Saver users

Role	Functionalities
User	• Sign up and log in. • Add, edit and delete food items. • Receive expiry alerts. • Get recipe suggestions. • Post donations and request food. • Search and filter donations. • Accept matches. • Share and rate recipes. • View rewards and volunteer hours.
Admin	• Log in to the admin account. • Review donation posts. • Approve or remove shared recipes.

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	• Manage user accounts (view, suspend or delete).

1.5 References

[1] IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications. New York, NY, USA: IEEE, 1998.

[2] K. E. Wiegers, “Software Requirements Specification Template,” Process Impact, 1999.

[3] I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson, 2016.

[4] United Nations Environment Programme, Food Waste Index Report 2024. Nairobi, Kenya: UNEP, 2024.

[5] Object Management Group, OMG Unified Modeling Language (UML), Version 2.5.1, Dec. 2017.

[6] K. Schwaber and J. Sutherland, The Scrum Guide, Nov. 2020.

[7] CS Year 5, SE Old Section, G1, “Software Project Management Plan for Food Saver,” Imam Abdulrahman Bin Faisal University, Jubail, Saudi Arabia, 2026.

2. Overall Description

2.1 Product Perspective

Food Saver is a new, self-contained product. It is not a replacement for an existing system and it is not part of a product family. The idea came from the problem of food waste at home and the lack of an easy way to share extra food with people nearby. Some existing apps focus only on tracking food or only on donation, while Food Saver brings tracking, alerts, recipes, donation and rewards together in one place.

The context model shows what lies outside the system boundaries and how Food Saver is related to the other systems around it. Food Saver is connected to seven systems: the user mobile app that users work with, the admin dashboard that the admin uses to manage the content, a database that stores all the data, a notification service that sends expiry alerts and match notices, a maps and location service that finds the distance between users, an email service that sends password reset links, and a volunteering platform that turns the points of users into volunteer hours. Figure 1 shows the context diagram of the system.

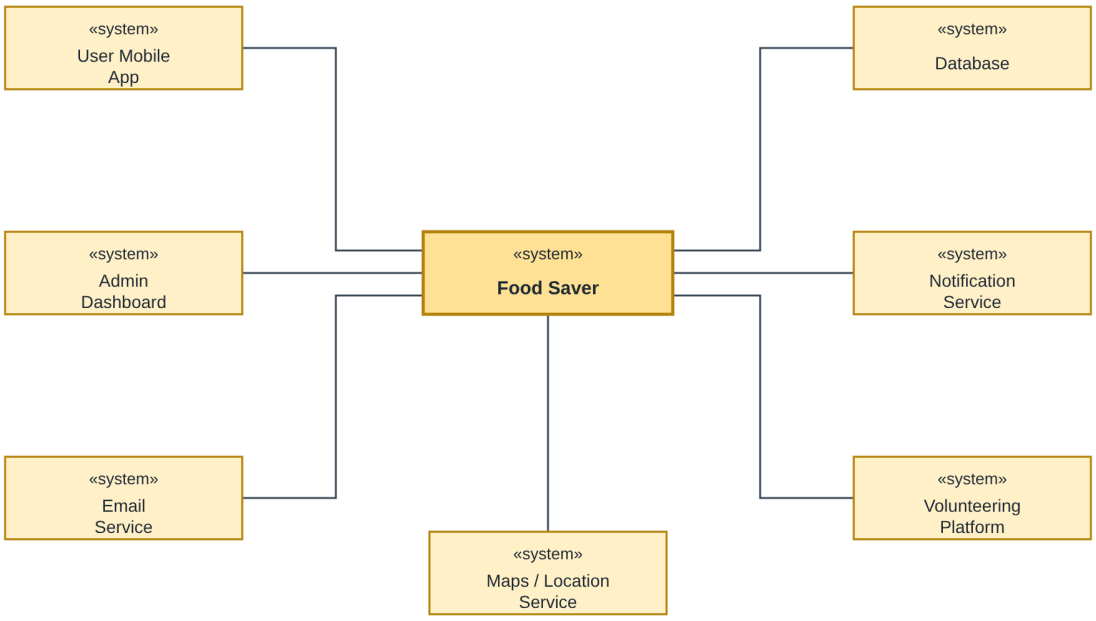


Figure 1: Context diagram of the Food Saver system

2.2 Product Functions

Food Saver is divided into the following main modules. Each module is explained in detail in Section 4, and the overall use case diagram is shown in Appendix B.

1. Sign Up and Log In

- New users create an account with their name, email, password and location.
- Registered users log in with their email and password and can reset a forgotten password.

2. Food Inventory

- Users add food items with name, category, quantity and expiry date.
- Users edit or delete items, and see them sorted by expiry date.

3. Expiry Alerts

- The system checks the expiry dates every day.
- Users receive an alert before an item expires and can choose when to be alerted.

4. Smart Recipe Suggestions

- The system suggests recipes based on the items in the inventory.
- Items that will expire soon are given priority in the suggestions.

5. Food Donation

- Users post extra food as a donation with its details and pickup area.
- Users who need food create a food request.

6. Search and Filter

- Users search the available donations by name.
- Users filter the results by food type, distance and expiry date.

7. Donation Matching

- The system matches donations with requests by location, food type and quantity.
- Both users are notified and the match is confirmed when it is accepted.

8. Recipe Community

- Users share their own recipes with other users.
- Users discover and rate shared recipes after the admin approves them.

9. Rewards

- Users earn points for every completed donation.
- Points are turned into volunteer hours through the volunteering platform.

10. Admin Management

- The admin reviews donation posts and approves shared recipes.
- The admin manages user accounts.

2.3 User Classes and Characteristics

Food Saver has two user classes:

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User: The user is anyone who uses Food Saver in their home, whether it is a student, a family member, or someone living alone. Users do not need any technical skills, just basic experience with smartphone apps. The same user can be a donor (while posting extra food) and a receiver (while requesting food). The users are the most important class, because most of the features are built for them. They are expected to use the application a lot, mostly for adding items and checking alerts.

Admin: The admin is a part of the Food Saver team and makes sure the content stays safe and appropriate. The admin has more privileges than a normal user, like reviewing donations, approving shared recipes and managing accounts. There are only a few admins, and they are expected to know the system and its rules well.

2.4 Operating Environment

Food Saver is a mobile application. It works in the following environment:

- Device: Smartphones with internet connection, location services (GPS) and touch screen.
- Operating System: Android 10 or higher and iOS 15 or higher.
- Database: a cloud database to store the user, food, recipe and donation data.
- Notifications: a push notification service for alerts and match notifications.
- Prototype: the current version is built in Figma and can be run using Figma on a web browser or on the Figma mobile app.

2.5 Design and Implementation Constraints

The following constraints limit the options available to the team while designing and building Food Saver. They are summarized in Table 3.

Table 3: Design and implementation constraints

ID	Constraint	Description
C1	Time	The project must be finished within one academic semester, following the schedule in the SPMP.
C2	Prototype	This version is an interactive prototype built in Figma, and no code is written.
C3	Sample data	Only sample data is used, so no real names, addresses or locations are stored.
C4	Team size	The project is done by a team of four students using Scrum.
C5	Language	The interface and documents are in English only in this version.
C6	Internet	Most features (alerts, donations, matching) need an internet connection.
C7	Standards	The documents follow IEEE standards and the diagrams follow UML 2.5.
C8	Privacy	The exact address of the donor shall only be shown after a donation is accepted.

2.6 User Documentation

The following documentation will be delivered with Food Saver:

- A short welcome guide (onboarding screens) that appears the first time the user opens the application and explains the main features.
- A Help and FAQ page inside the application that answers common questions, such as how to add an item or how points are calculated.
- A short user guide in PDF format that explains the main tasks step by step with screenshots of the prototype.
- An admin guide that explains how to review donations, approve recipes and manage users.

2.7 Assumptions and Dependencies

- Users are assumed to enter correct expiry dates for their food items.
- Users are assumed to allow location access, because donation matching depends on it.
- Users are assumed to have a smartphone with an internet connection.
- Donors are responsible for the quality and safety of the food they donate.
- The rewards feature depends on an external volunteering platform that accepts donation records and returns volunteer hours (TBD-1).
- The alerts depend on a push notification service being available.
- The recipe suggestions depend on a recipe database that contains enough recipes for common ingredients.

3. External Interface Requirements

3.1 User Interfaces

The user interface is designed to be simple and clear for all ages. All screens use the same green color theme, the same fonts and a bottom navigation bar with five tabs: Home, Recipes, Donate, Community and Profile. Error messages appear in red under the field that has the problem, and success messages appear as a short message at the bottom of the screen. The following tables describe the main components of each screen. Screenshots of the screens can be found in the Figma prototype.

Table 4: Sign Up screen components

Component	Description
Name field	Text field for the full name of the user.
Email field	Text field for the email address.
Password field	Hidden text field for the password (at least 8 characters).
Location field	Field to choose the city or allow location access.
Sign Up button	Creates the account after the data is checked.
Log In link	Takes the user to the log in screen if they already have an account.

Table 5: Log In screen components

Component	Description
Email field	Text field for the registered email.
Password field	Hidden text field for the password.
Log In button	Checks the data and opens the home screen.
Forgot Password link	Opens the reset password screen.
Sign Up link	Takes a new user to the sign up screen.

Table 6: Home (Food Inventory) screen components

Component	Description
Item list	Shows the food items sorted by expiry date, with a colored label for items that expire soon.
Add button (+)	Opens the add food item screen.
Category tabs	Filter the list by category, such as dairy, vegetables or bread.
Alerts icon	Opens the alerts screen and shows the number of new alerts.
Navigation bar	Moves between Home, Recipes, Donate, Community and Profile.

Table 7: Add / Edit Food Item screen components

Component	Description
Name field	Text field for the item name.
Category list	Drop-down list to choose the category.
Quantity field	Number field for the quantity.
Expiry date picker	Calendar to choose the expiry date.
Save button	Saves the item in the inventory.

Delete button	Shown only when editing; removes the item after confirmation.

Table 8: Alerts screen components

Component	Description
Alert list	Shows the items that will expire soon, with the number of days left.
Recipe shortcut	Opens recipe suggestions for the selected item.
Donate shortcut	Opens the donation form with the item details filled in.
Alert settings	Lets the user choose how many days before expiry to be alerted.

Table 9: Recipe Suggestions screen components

Component	Description
Recipe cards	Show the recipe title, picture, time and the items it uses from the inventory.
Expiring soon label	Marks recipes that use items close to their expiry date.
Recipe details	Show the ingredients and steps of the selected recipe.

Table 10: Donate / Request Food screen components

Component	Description
Donate and Request tabs	Switch between posting a donation and creating a food request.
Food details fields	Food type, quantity and expiry date.
Pickup area field	General area of the donor (the exact address is hidden).
Post button	Posts the donation or request.

Table 11: Donations Search screen components

Component	Description
Search bar	Text field to search donations by name.
Filter button	Opens filters for food type, distance and expiry date.
Results list	Shows matching donations with the food type, quantity and distance.
Donation details	Show the full details of the selected donation and a Request button.

Table 12: Match screen components

Component	Description
Match card	Shows the matched donation or request with the distance and food details.
Accept button	Confirms the match.
Decline button	Rejects the match so the system can look for another one.
Pickup details	Show the exact address and time after the match is accepted.

Table 13: Recipe Community screen components

Component	Description
Shared recipe list	Shows approved recipes shared by users.
Share Recipe button	Opens a form to add a title, ingredients, steps and a picture.
Rating stars	Let the user rate a recipe from 1 to 5.

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Table 14: Rewards screen components

Component	Description
Points counter	Shows the total points of the user.
Volunteer hours	Shows the hours confirmed by the volunteering platform.
History list	Shows the completed donations and the points earned for each one.

Table 15: Profile screen components

Component	Description
User information	Shows the name, email and location.
Edit Profile button	Lets the user update their information.
Settings	Notification and privacy settings.
Log Out button	Ends the session and returns to the log in screen.

3.2 Hardware Interfaces

Food Saver does not need any special hardware. It runs on normal smartphones and uses the following parts of the device:

- Touch screen: used for all inputs, such as typing, tapping and scrolling.
- GPS: used to get the location of the user, which is needed for donation matching and distance filters.
- Network (Wi-Fi or mobile data): used to send and receive data from the database and the notification service.
- Storage: used to keep a small copy of the inventory so it can be viewed without the internet.

3.3 Software Interfaces

Food Saver connects with the software components shown in Table 16.

Table 16: Software interfaces

Component	Type	Data Shared	Purpose
Database	Cloud database	User accounts, food items, recipes, donations, requests, matches and points.	Stores all the data of the application and returns it when needed.
Notification Service	Push notification service	Alert messages and match notices.	Sends expiry alerts and match notices to the user's phone.
Volunteering Platform	External platform (TBD-1)	Completed donation records (out) and confirmed volunteer hours (in).	Turns the points of the user into official volunteer hours.
Maps / Location Service	Device location service	Coordinates of the user.	Calculates the distance between donors and receivers.

Operating System	Android 10+ / iOS 15+	Device permissions and notifications.	Runs the application and gives access to GPS and notifications.
User Mobile App	Mobile application (Android / iOS)	User inputs (in) and screens, alerts and results (out).	Lets users use all the features of Food Saver on their phones.
Admin Dashboard	Web dashboard	Donation posts, shared recipes and user accounts.	Lets the admin review posts, approve recipes and manage users.
Email Service	Email service	Password reset links.	Sends password reset links to the registered email of the user.

3.4 Communications Interfaces

- All data between the application and the server shall be sent using HTTPS to keep it secure.
- Expiry alerts and match notices shall be sent as push notifications.
- Password reset links shall be sent to the registered email address of the user.
- Data that is changed while the user is offline shall be synchronized when the connection comes back.

4. System Features

This section presents the main functional requirements of Food Saver, organized by system feature. Each feature has a description and priority, a use case diagram, a use case table and a list of functional requirements.

4.1 Sign Up

4.1.1 Description and Priority

Sign up allows new users to create an account in Food Saver so they can use all of its features. The user enters basic information, and the system checks it before saving the account. Figure 2 shows the use case diagram of this feature. The priority of this feature is high. Benefit: 9 - Penalty: 8 - Cost: 4 - Risk: 3.

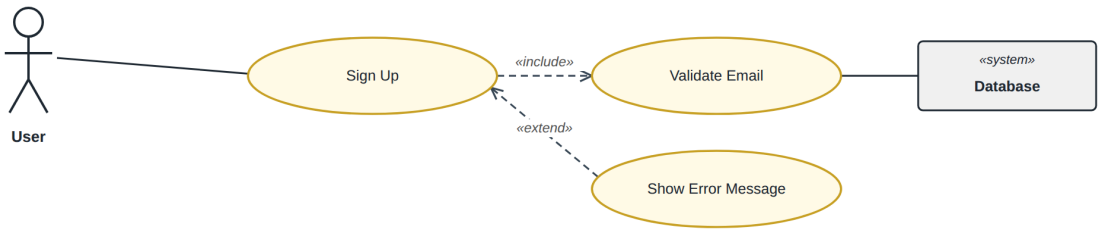


Figure 2: Use case diagram of the sign up feature

4.1.2 Stimulus/Response Sequences

Table 17: Use case of sign up

Food Saver: Sign Up	
Actors	User, Database
Description	A new user creates an account by entering a name, email, password and location. The system checks the data and saves the new account in the database.
Data	Name, email, password and location of the user.
Stimulus	The user selects the “Sign Up” button.
Response	The account is created, a success message is shown and the user is taken to the home screen.
Comments	The email must be valid and not used before, and the password must be at least 8 characters.

4.1.3 Functional Requirements

REQ-1: The system allows a new user to create an account by entering a name, email, password and location.

REQ-2: The system checks that the email is in a valid format and is not already registered.

REQ-3: The system requires the password to be at least 8 characters long.

REQ-4: The system displays an error message when any required field is empty or invalid.

REQ-5: The system saves the new account in the database and displays a success message.

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4.2 Log In

4.2.1 Description and Priority

Log in allows registered users and admins to enter their accounts using their email and password. If a user forgets the password, they can reset it through their email. Figure 3 shows the use case diagram of this feature. The priority of this feature is high. Benefit: 9 - Penalty: 8 - Cost: 3 - Risk: 3.

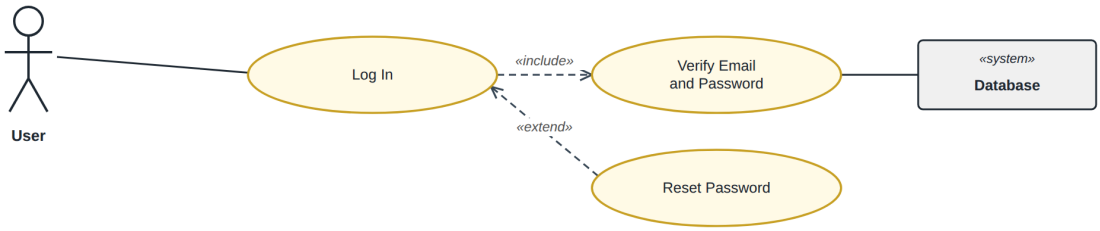


Figure 3: Use case diagram of the log in feature

4.2.2 Stimulus/Response Sequences

Table 18: Use case of log in

Food Saver: Log In	
Actors	User, Admin, Database
Description	A registered user or admin enters the email and password to access the account. If the password is forgotten, a reset link is sent to the registered email.
Data	Email and password.
Stimulus	The user selects the “Log In” button.
Response	The home screen (or the admin dashboard) is shown, or an error message if the email or password is wrong.
Comments	The account is locked for 15 minutes after 5 failed log in attempts.

4.2.3 Functional Requirements

REQ-6: The system allows registered users to log in using their email and password.

REQ-7: The system verifies the email and password before giving access to the account.

REQ-8: The system displays an error message when the email or password is incorrect.

REQ-9: The system allows the user to reset the password through a link sent to the registered email.

REQ-10: The system locks the account for 15 minutes after 5 failed log in attempts in a row.

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4.3 Food Inventory

4.3.1 Description and Priority

The food inventory is the main feature of Food Saver. It allows the user to keep a list of the food they have at home with the quantity and expiry date of each item. The other features, such as alerts and recipe suggestions, depend on this list. Figure 4 shows the use case diagram of this feature. The priority of this feature is high. Benefit: 9 - Penalty: 9 - Cost: 6 - Risk: 4.

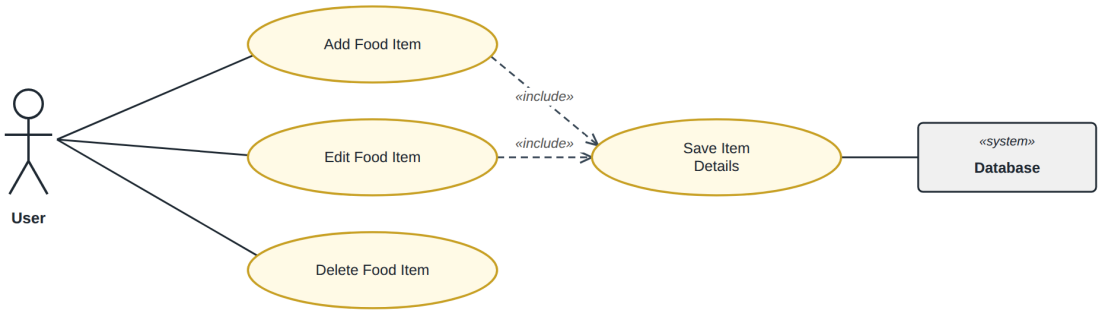


Figure 4: Use case diagram of the food inventory feature

4.3.2 Stimulus/Response Sequences

Table 19: Use case of add food item

Food Saver: Add Food Item	
Actors	User, Database
Description	The user adds a food item to the inventory with its details. The user can also edit or delete items later, and the list is sorted by expiry date.
Data	Item name, category, quantity and expiry date.
Stimulus	The user taps the add button (+) on the home screen.
Response	The item is saved and shown in the inventory list, sorted by expiry date.
Comments	An expiry date in the past is not accepted.

4.3.3 Functional Requirements

REQ-11: The system shall allow the user to add a food item with name, category, quantity and expiry date.

REQ-12: The system shall allow the user to edit the details of any food item.

REQ-13: The system shall ask for confirmation before deleting a food item.

REQ-14: The system shall display the inventory sorted by expiry date, from the closest to the furthest.

REQ-15: The system shall mark items that expire within 3 days with a colored label.

REQ-16: The system shall not accept an expiry date that is in the past when adding a new item.

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4.4 Expiry Alerts

4.4.1 Description and Priority

Expiry alerts remind the user about food items before they expire, so the user can use them, cook them or donate them in time. The user can choose when to receive the alert. Figure 5 shows the use case diagram of this feature. The priority of this feature is high. Benefit: 9 - Penalty: 8 - Cost: 5 - Risk: 4.

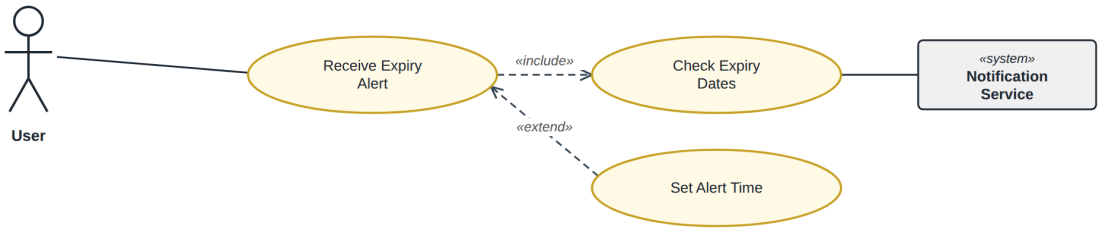


Figure 5: Use case diagram of the expiry alerts feature

4.4.2 Stimulus/Response Sequences

Table 20: Use case of receive expiry alert

Food Saver: Receive Expiry Alert	
Actors	User, Notification Service
Description	The system checks the expiry dates of the items every day and alerts the user before an item expires, so it can be used, cooked or donated in time.
Data	Item name, expiry date and the number of days left.
Stimulus	An item reaches the alert time chosen by the user (2 days before expiry by default).
Response	An alert is sent to the user's phone and shown in the alerts list, with options to view recipes or donate the item.
Comments	The user can change the alert time from 1 to 7 days before expiry.

4.4.3 Functional Requirements

REQ-17: The system shall check the expiry dates of all items in the inventory once every day.

REQ-18: The system shall send an alert 2 days before an item expires by default.

REQ-19: The system shall allow the user to change the alert time from 1 to 7 days before expiry.

REQ-20: The system shall show a list of all alerts inside the application.

REQ-21: The system shall give the user the option to view recipes or donate the item from the alert.

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4.5 Smart Recipe Suggestions

4.5.1 Description and Priority

Smart recipe suggestions help the user cook meals using the food they already have. The system reads the inventory and suggests recipes, giving priority to the items that will expire soon. Figure 6 shows the use case diagram of this feature. The priority of this feature is medium. Benefit: 8 - Penalty: 6 - Cost: 7 - Risk: 5.

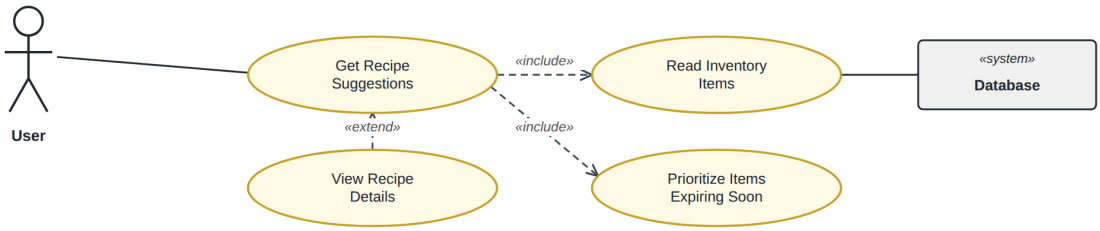


Figure 6: Use case diagram of the smart recipe suggestions feature

4.5.2 Stimulus/Response Sequences

Table 21: Use case of get recipe suggestions

Food Saver: Get Recipe Suggestions	
Actors	User, Database
Description	The system reads the items in the user's inventory and suggests recipes that use them, starting with the items that will expire soon.
Data	Inventory items, expiry dates and recipe ingredients.
Stimulus	The user opens the Recipes tab or selects a recipe shortcut from an alert.
Response	A list of suitable recipes is shown, with the available and missing ingredients of each recipe.
Comments	If no recipe matches the inventory, popular recipes are suggested instead.

4.5.3 Functional Requirements

REQ-22: The system shall suggest recipes based on the items in the user's inventory.

REQ-23: The system shall show recipes that use items expiring soon at the top of the list.

REQ-24: The system shall show for each recipe which ingredients are available in the inventory and which are missing.

REQ-25: The system shall display the ingredients and steps of the recipe the user selects.

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4.6 Food Donation

4.6.1 Description and Priority

Food donation allows users to post extra food that they will not use, and allows other users to request food they need. The admin reviews the posts to make sure they follow the rules. Figure 7 shows the use case diagram of this feature. The priority of this feature is high. Benefit: 9 - Penalty: 8 - Cost: 6 - Risk: 5.

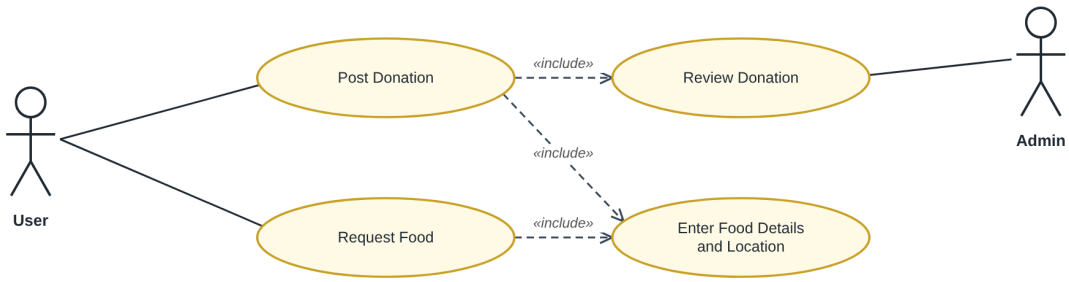


Figure 7: Use case diagram of the food donation feature

4.6.2 Stimulus/Response Sequences

Table 22: Use case of post donation

Food Saver: Post Donation	
Actors	User, Admin
Description	A user posts extra food as a donation so people nearby can request it. Users who need food can also create a food request. The admin reviews the posts.
Data	Food type, quantity, expiry date and pickup area.
Stimulus	The user selects “Donate” from the navigation bar or from an alert.
Response	The donation is posted, shown to other users and sent to the admin for review.
Comments	Expired food cannot be donated, and only the general area of the donor is shown until a match is accepted.

4.6.3 Functional Requirements

REQ-26: The system shall allow the user to post a donation with food type, quantity, expiry date and pickup area.

REQ-27: The system shall allow the user to create a food request with food type, quantity and location.

REQ-28: The system shall not accept a donation of food that has already expired.

REQ-29: The system shall show only the general area of the donor until the donation is accepted.

REQ-30: The system shall allow the user to cancel a donation or a request before it is matched.

REQ-31: The system shall allow the admin to review and remove donations that break the rules.

4.7 Search and Filter

4.7.1 Description and Priority

Search and filter help the user find available donations quickly. The user can search by name and filter the results by food type, distance and expiry date. Figure 8 shows the use case diagram of this feature. The priority of this feature is medium. Benefit: 7 - Penalty: 5 - Cost: 4 - Risk: 2.

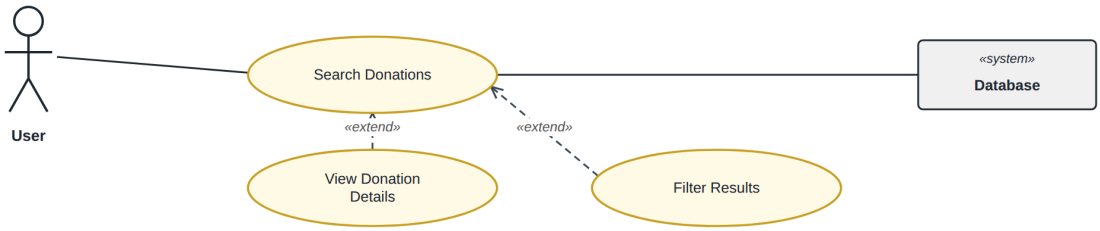


Figure 8: Use case diagram of the search and filter feature

4.7.2 Stimulus/Response Sequences

Table 23: Use case of search donations

Food Saver: Search Donations	
Actors	User, Database
Description	The user searches the available donations by food name and filters the results by food type, distance and expiry date.
Data	Search word and filter values (food type, distance and expiry date).
Stimulus	The user types in the search bar on the donations screen.
Response	A list of matching donations is shown, sorted by distance from the user.
Comments	A “no results” message is shown when no donation matches the search.

4.7.3 Functional Requirements

REQ-32: The system shall allow the user to search the available donations by food name.

REQ-33: The system shall allow the user to filter the results by food type, distance and expiry date.

REQ-34: The system shall display the results sorted by distance from the user by default.

REQ-35: The system shall display a message when no donation matches the search.

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4.8 Donation Matching

4.8.1 Description and Priority

Donation matching connects donations with food requests automatically. The system compares the location, food type and quantity, then notifies both users so they can accept the match. Figure 9 shows the use case diagram of this feature. The priority of this feature is high. Benefit: 8 - Penalty: 7 - Cost: 7 - Risk: 6.

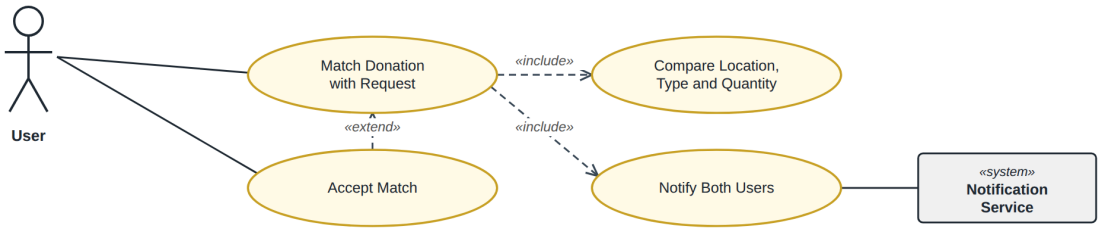


Figure 9: Use case diagram of the donation matching feature

4.8.2 Stimulus/Response Sequences

Table 24: Use case of match donation with request

Food Saver: Match Donation with Request	
Actors	User, Notification Service
Description	The system compares new donations and requests by location, food type and quantity, and creates a match with the closest suitable user. Both users are notified and must accept the match.
Data	Location, food type and quantity of the donation and the request.
Stimulus	A new donation or a new food request is posted.
Response	Both users receive a match notice, and after both accept, the exact pickup address is shown.
Comments	Users are matched only within a maximum distance (TBD-2). If one user declines, the system looks for another match.

4.8.3 Functional Requirements

REQ-36: The system shall match donations with requests based on location, food type and quantity.

REQ-37: The system shall only match users within a maximum distance (TBD-2).

REQ-38: The system shall notify both users when a match is found.

REQ-39: The system shall confirm the match only after both users accept it.

REQ-40: The system shall show the exact pickup address only after the match is accepted.

REQ-41: The system shall look for another match if one of the users declines.

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4.9 Recipe Community

4.9.1 Description and Priority

The recipe community allows users to share their own recipes, especially recipes that use leftovers, and to discover and rate recipes shared by others. Shared recipes are approved by the admin before they appear. Figure 10 shows the use case diagram of this feature. The priority of this feature is low. Benefit: 6 - Penalty: 4 - Cost: 5 - Risk: 3.

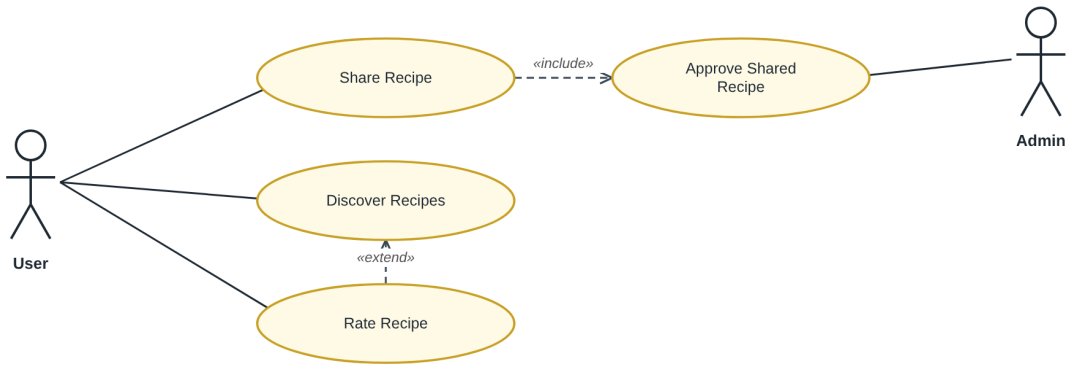


Figure 10: Use case diagram of the recipe community feature

4.9.2 Stimulus/Response Sequences

Table 25: Use case of share recipe

Food Saver: Share Recipe	
Actors	User, Admin
Description	A user shares a recipe with the community. The admin approves it before it is published, and other users can rate it.
Data	Recipe title, ingredients, steps, picture and rating.
Stimulus	The user selects “Share Recipe” in the Community tab.
Response	The recipe is sent to the admin and published in the community after approval.
Comments	Users cannot rate their own recipes.

4.9.3 Functional Requirements

REQ-42: The system shall allow the user to share a recipe with a title, ingredients, steps and a picture.

REQ-43: The system shall publish a shared recipe only after the admin approves it.

REQ-44: The system shall allow users to browse and search shared recipes.

REQ-45: The system shall allow the user to rate a recipe from 1 to 5 stars.

REQ-46: The system shall not allow users to rate their own recipes.

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4.10 Rewards

4.10.1 Description and Priority

Rewards encourage users to keep donating. The user earns points for each completed donation, and the points are turned into volunteer hours through a volunteering platform. In this version, the rewards are shown using sample data. Figure 11 shows the use case diagram of this feature. The priority of this feature is low. Benefit: 6 - Penalty: 3 - Cost: 6 - Risk: 5.

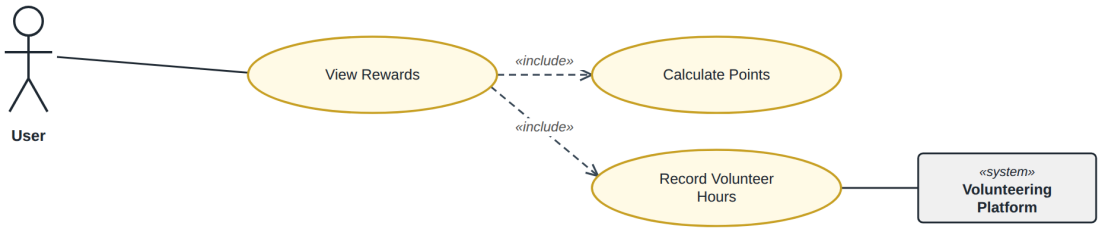


Figure 11: Use case diagram of the rewards feature

4.10.2 Stimulus/Response Sequences

Table 26: Use case of view rewards

Food Saver: View Rewards	
Actors	User, Volunteering Platform
Description	The user views the points earned from completed donations and the volunteer hours confirmed by the volunteering platform.
Data	Points, completed donations and volunteer hours.
Stimulus	The user opens the Rewards screen from the profile.
Response	The total points, the volunteer hours and the donation history are shown.
Comments	The rate used to turn points into hours is not decided yet (TBD-3). This version shows the rewards using sample data.

4.10.3 Functional Requirements

REQ-47: The system shall add points to the user's account after each completed donation.

REQ-48: The system shall turn the points into volunteer hours using a fixed rate (TBD-3).

REQ-49: The system shall send the completed donations to the volunteering platform to confirm the volunteer hours.

REQ-50: The system shall display the total points, the volunteer hours and the donation history.

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The application shall open any main screen within 2 seconds on a normal 4G connection.
- The system shall return search results within 3 seconds.
- Expiry alerts shall be sent within 5 minutes of the alert time.
- The system shall find a match within 1 minute after a donation or request is posted, if a suitable one exists.
- The system shall support at least 1,000 users using it at the same time without slowing down.

5.2 Safety Requirements

- The system shall not allow the donation of food that has passed its expiry date.
- The system shall show a short food safety reminder to the donor before posting a donation (for example, keep cold food in the fridge until pickup).
- The exact address of the donor shall be hidden until the match is accepted, to protect the safety of the users.
- The system shall allow users to report a donation or a user that seems unsafe, and the admin shall review the report.

5.3 Security Requirements

- Passwords shall be stored in encrypted (hashed) form and never shown in plain text.
- All data shall be sent between the application and the server using HTTPS.
- Only the admin shall be able to review donations, approve recipes and manage user accounts.
- The account shall be locked for 15 minutes after 5 failed log in attempts.
- The location of the user shall only be used for matching and distance, and shall not be shared with other users before a match is accepted.
- The system shall follow the Saudi Personal Data Protection Law (PDPL) when real user data is used.

5.4 Software Quality Attributes

Usability: A new user shall be able to add a food item within 1 minute without help, and the main features shall be reachable from the navigation bar in at most 2 taps.

Availability: The system shall be available 99% of the time, except during planned maintenance.

Reliability: The system shall not lose any saved data, and alerts shall be delivered correctly for at least 99% of items.

Maintainability: The system shall be divided into separate modules (as in Section 2.2) so each module can be updated without affecting the others.

Portability: The application shall run on both Android and iOS with the same features and design.

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Scalability: The system shall be able to grow to more users and more cities without changing its design.

5.5 Business Rules

- Only registered users can make donations, request food, share recipes or earn points.
- Food can only be donated if the expiry date is not less than one day after the date of posting.
- Each donation can be matched with only one request.
- Points are added only after a donation is marked as completed by both users.
- Users cannot rate their own recipes.
- Only the admin can approve or remove shared recipes and donation posts.
- The admin can suspend an account for breaking the rules more than once.

6. Other Requirements

Database Requirements: The system shall store all data in a cloud database. Each user, food item, recipe, post and match shall have a unique ID, and the relations between them follow the class diagram in Appendix B.

Language Requirements: The application is provided in English in this version. Arabic support is planned for a future version (TBD-4).

Backup Requirements: The database shall be backed up on a daily basis and the backup shall be retained for a minimum of 30 days.

Synchronization: Changes made while the user is offline shall be stored on the device and synchronized automatically when the connection is re-established.

Legal Requirements: The application shall obtain the user's consent before it accesses their location or sends notifications, and a privacy policy shall be displayed at sign up.

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Appendix A: Glossary

Table 27: Glossary

Term	Definition
Admin	A team member who reviews donations, approves recipes and manages users.
Donation	Extra food that a user offers to other users for free.
Expiry Alert	A notice sent to the user before a food item expires.
Figma	A design tool used to build the interactive prototype of the application.
Food Inventory	The list of food items that a user has at home.
Food Request	A post created by a user who needs a certain type of food.
HTTPS	Hypertext Transfer Protocol Secure, used to send data safely over the internet.
IEEE	Institute of Electrical and Electronics Engineers.
Match	A link created by the system between a donation and a food request.
PDPL	Personal Data Protection Law of Saudi Arabia.
Prototype	An early interactive model of the application that shows how it works using sample data.
REQ	Short for requirement; used to number the functional requirements.
Scrum	An agile framework used by the team to manage the project in short sprints.
SPMP	Software Project Management Plan.
SRS	Software Requirements Specification.
TBD	To Be Determined; information that is not decided yet.
UML	Unified Modeling Language, used to draw the diagrams in this document.
Volunteer Hours	Official hours given by the volunteering platform in return for the user's donations.

Appendix B: Analysis Models

This appendix includes the main analysis models of Food Saver. Figure 12 shows the overall use case diagram, which brings all the features together, and Figure 13 shows the class diagram of the system with the main classes, their attributes and operations, and the relations between them. To keep the diagram simple, similar classes are combined: users and admins are both stored in the User class with a role, and donations and food requests are both stored in the Post class with a type.



Figure 12: Overall use case diagram of Food Saver

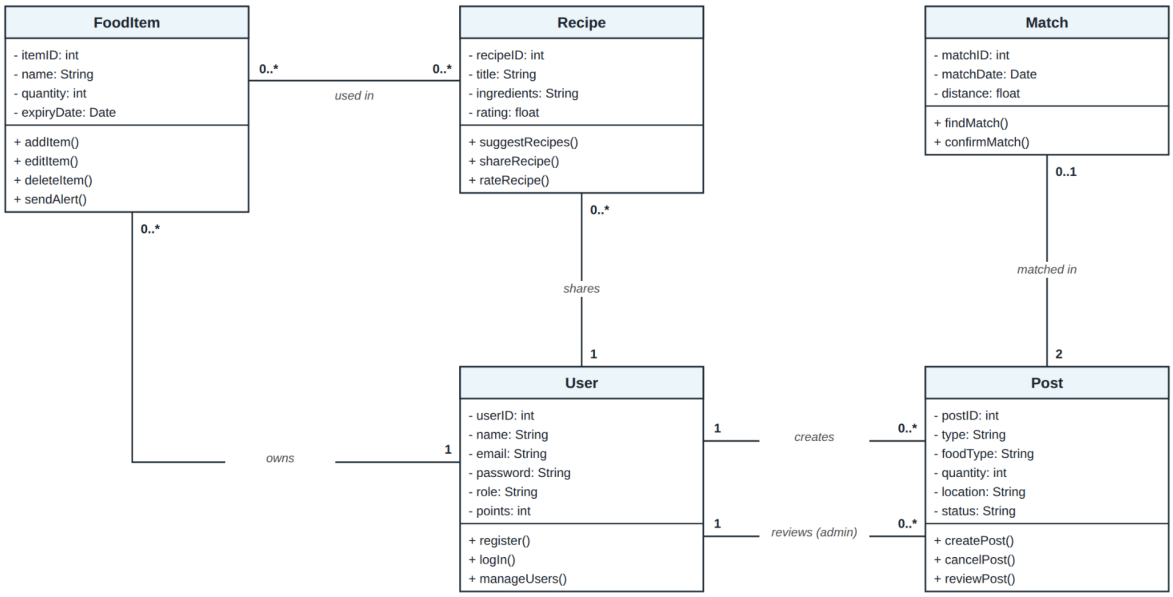


Figure 13: Class diagram of Food Saver

Appendix C: To Be Determined List

- TBD-1: The name of the volunteering platform that Food Saver will connect to, and the way the data will be exchanged with it.
- TBD-2: The maximum distance between the donor and the receiver for a match (for example 5 or 10 km).
- TBD-3: The rate used to turn points into volunteer hours.
- TBD-4: The date for adding Arabic language support.